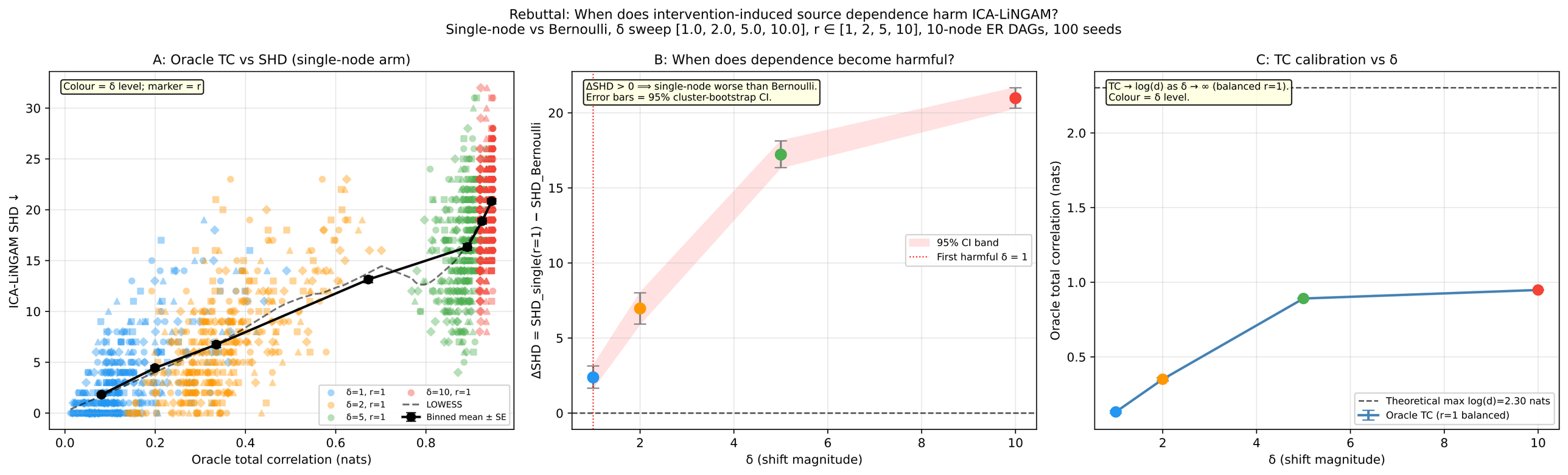
Rebuttal: When does intervention-induced source dependence harm ICA-LiNGAM?
Single-node vs Bernoulli, δ sweep [1.0, 2.0, 5.0, 10.0], r ∈ [1, 2, 5, 10], 10-node ER DAGs, 100 seeds
A: Oracle TC vs SHD (single-node arm)
Colour = δ level; marker = r
ICA-LiNGAM SHD ↓
Oracle total correlation (nats)
δ=1, r=1
δ=2, r=1
δ=5, r=1
δ=10, r=1
LOWESS
Binned mean ± SE
B: When does dependence become harmful?
ΔSHD > 0 ⟹ single-node worse than Bernoulli.
Error bars = 95% cluster-bootstrap CI.
ΔSHD = SHD_single(r=1) − SHD_Bernoulli
δ (shift magnitude)
95% CI band
First harmful δ = 1
C: TC calibration vs δ
TC → log(d) as δ → ∞ (balanced r=1).
Colour = δ level.
Oracle total correlation (nats)
δ (shift magnitude)
Theoretical max log(d)=2.30 nats
Oracle TC (r=1 balanced)